N-Mixture Model and More

Objectives

- N-mixture model
- Occurrence Probability and Abundance
- Integrated Data Modeling

A new model

- Royle 2004
  - Bayesian Analysis of Spatially Structured Population Dynamics
- Another way to estimate abundance with imperfect detection
  - mark recapture
  - removal sampling
  - distance sampling
  - double observer

N-mixture model

- Fundamentally a 'count model' that corrects for detection probability
- "Capture–recapture studies can be difficult to implement" (Royle, 2004)
  - Maybe more useful for sparse data (few sites, few occasions, small detection probability, some zeros)
- **Data Inputs**: Replicated spatial counts
- **Abundance**: State process is a Poisson distribution
- **Detection**: Observation process is a Binomial distribution

N-mixture model

$y_{i,t}$ = number of individuals observed at site $i = 1...R$ during $t = 1...T$ sampling occasions.

. . .

$$y_{i,t} \sim \text{Binomial}(N_i, p)$$
$$N_i \sim \text{Poisson}(\lambda)$$

. . .

- N_i = size of population at site i
- $\sum_{i=1}^{R} N_i$ = total population size

. . .

Priors:

- $\log(\lambda) \sim \text{Normal}(0, 3)$
- OR, $\lambda \sim \text{Gamma}(0.01, 0.01)$
- $\text{logit}(p) \sim \text{Logistic}(0, 1)$
- OR, $p \sim \text{Beta}(1, 1)$

N Mixture Model

Assumptions:

1) Poisson and binomial distributions are true descriptions of state and observation processes; i.e., variation in abundance can be described by the assumed distribution (e.g., Poisson).
2) Abundance at each site i is random and independent of abundance at all other sites;
3) Population is closed among surveys
4) Observers do not double count individuals
5) All individuals have the same detection probability or otherwise modeled

Data

Site	N	Visit 1	Visit 2	Visit 3	Visit 4	Visit 5
1	6	2	3	3	5	3
2	2	0	2	1	1	1
3	4	0	0	3	3	2
4	1	0	1	0	0	1

N-mixture extensions

$y_{i,t}$ = number of individuals observed at site $i = 1...R$ during $t = 1...T$ sampling occasions.

$$\begin{aligned} y_{i,t} &\sim \text{Binomial}(N_i, p_{i,t}) \\ N_i &\sim \text{Poisson}(\lambda_i) \end{aligned}$$

. . .

$$\begin{aligned} \text{logit}(p_{i,t}) &= \beta_0 + \beta_1 \times x_{i,t} \\ \log(\lambda_i) &= \alpha_0 + \alpha_1 \times w_i + \epsilon_i \\ \epsilon_i &\sim \text{Normal}(0, \sigma^2) \end{aligned}$$

More extensions

- Other distributions
  - Negative-binomial
  - Zero-inflated Poisson
  - Multinomial
- Integrate known date or distance information
- Multiple states, ages, or sexes
- Spatial movement, e.g., temporary emigration

Implementation

- Maximum likelihood
  - Unmarked pacakge; pcount function
- Bayesian
  - ubms package
  - JAGS, nimble, INLA, stan

Concerns

Parameter Identifiability: "A model is identifiable if there is only one set of parameter values that result in a single model output". (Cole book)

. . .

Data Identifiability: a condition of the unique dataset that does not allow the model to estimate a unique set of parameters to be output

. . .

Table 2.8: Parameter estimates from fitting the N-Mixture model

Method	(a) λ	(a) p	(b) λ	(b) p
PRESENCE	45.26	0.05	7.98	0.29
unmarked $K = 100$	31.19	0.07	8.00	0.29
unmarked $K = 500$	68.72	0.03	8.00	0.29
unmarked $K = 1000$	862.64	0.003	8.00	0.29
Dennis et al. (2015)	5.35×10^{10}	4.30×10^{-11}	7.98	0.29

Concerns

Link et al. 2018

Barker et al. 2017

- 2% double counting can lead to ~20% bias
- small unmodelled heterogenity in p can lead to 20% bias
- small distributional assumption violation leads to CI coverage of 17%

Concerns

Link et al. 2018

Barker et al. 2017

- "fundamentally count data under imperfect detection can only be reliably used as indices." (i.e., relative abundance, not absolute abundance)

. . .

… this is because without auxiliary information about p, count data cannot distinguish between an N-mixture model and other possible models (GLM).

Concerns

Kery, 2017

- Testing concern with 137 bird datasets
- Small sample size caused issues (<267 sites)
- Covariates did not affect identifiabiltiy
- No identifiability issues for Poisson and ZIP distributional assumptions
- 25% of datasets had issues with Negative Binomial distribution

N-mixture model

Go to code….

```
model {

  # Priors
  lambda ~ dgamma(0.01,0.01)
  p ~ dbeta(1,1)

  # Likelihood portion for data set 1: counts
  for (i in 1:nsites1){
    N[i] ~ dpois(lambda)
    for (j in 1:noccs){
      y[i,j] ~ dbinom(p,N[i])
    }
  }

}
```

Occurrence Probability and Expected Abundance

ψ = probability of occurrence

λ = mean count/abundance

. . .

$$\psi = P(Y = 1)$$

. . .

$$\psi = P(Y = 1) = P(N > 0)$$

. . .

$$\psi = P(Y = 1) = P(N > 0) = 1 - e^{-\lambda}$$

. . .

```
lambda = 2
psi = 1 - exp(-lambda)
psi
```

```
[1] 0.8646647
```

. . .

The Probability of observing > 0 counts, given lambda $= 2$

```
ppois(0,lambda=2,lower.tail = FALSE)
```

```
[1] 0.8646647
```

More to the idea

Starting with:

$$\psi = 1 - e^{-\lambda}$$

. . .

Lets rearrange:

$$e^{-\lambda} = 1 - \psi$$

. . .

$$\log(e^{-\lambda}) = \log(1 - \psi)$$

. . .

$$-\lambda = \log(1 - \psi)$$

. . .

$$\lambda = -\log(1 - \psi)$$

. . .

$$\log(\lambda) = \log(-\log(1 - \psi))$$

. . .

The right side is called the 'complimentary log-log link function'.

The end result

$$\log(\lambda) = \log(-\log(1-\psi))$$

OR

$$\log(-\log(1-\psi)) = \log(\lambda)$$

. . .

OR

$$\text{cloglog}(\psi) = \log(\lambda)$$

. . .

OR

$$\text{cloglog}(\psi) = \log(\beta_0 + \beta_1 \times x_1)$$

The model all together

y_i are the presence and absences of the species at site i...

$$y_i \sim \text{Bernoulli}(\psi_i)\text{cloglog}(\psi_i) = \beta_0 + \beta_1 \times x_i$$

Presence/Absence Model

Go to code....

```
model {

  # Priors
  beta0 ~ dunif(-10, 10)
  beta1 ~ dnorm(0, 1/5^2)

  # Likelihood portion for data set 3: presence/absence
  for (k in 1:nsites3){
    y[k] ~ dbern(psi[k])
    cloglog(psi[k]) <- beta0 + beta1 * selev3[k]
  }

}
```

Integrated Data Model

- a hierarchical model
- separate response variables where at least one parameter is shared among at least 2 response variables
- "It just makes sense to use all the information that is available in some estimation task" (Kery and Kellner, 2024)

Integrated Data Model

- **Names**: integrated model, data fusion, data assimilation
- specific types:
  - integrated population models
  - integrated species distribution models
  - integrated community model
  - integrate dynamic N-mixture model

Integrated Data Model

"datastream specific observation submodels that inform a common ecological model" Gilbert et al. 2021

Integrated Species Distribution Model

In chapter 20 of Kery and Kellner, 2024 they introduce three data sets.

1. Counts of swifts (y_1)
2. Counts of swifts, but the "lazy" birdwatchers didn't record the zero counts (y_2)
3. Observers recorded whether swifts were present (1) or not (0) (y_3)

Integrated Species Distribution Model

We could fit each separately, but…

. . .

Or, could fit a model that leverages all three data sets, each with their own observation model but a single joint 'process/state' model.

Integrated Data Model

Sub model 1

$$y_{1,i} \sim \text{Poisson}(\lambda)$$

Sub model 2

$$y_{2,i} \sim \text{ZT Poisson}(\lambda)$$

Sub model 3

$$y_{3,i} \sim \text{Binomial}(\psi)$$

Integrated Data Model

Sub model 1

$$\begin{aligned} y_{1,i} &\sim \text{Poisson}(\lambda_i) \\ \log(\lambda_i) &= \beta_0 + \beta_1 \times x_i \end{aligned}$$

Sub model 2

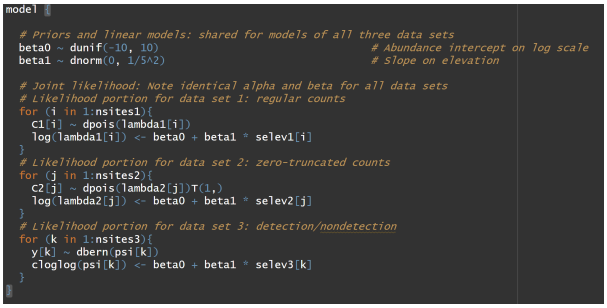
$$\begin{aligned} y_{2,i} &\sim \text{ZT Poisson}(\lambda_i) \\ \log(\lambda_i) &= \beta_0 + \beta_1 \times x_i \end{aligned}$$

Sub model 3

$$\begin{aligned} y_{3,i} &\sim \text{Binomial}(\psi_i) \\ \text{cloglog}(\psi) &= \beta_0 + \beta_1 \times x_i \end{aligned}$$

Integrated Data Model

Go to code...



```
model {

  # Priors and linear models: shared for models of all three data sets
  beta0 ~ dunif(-10, 10)                              # Abundance intercept on log scale
  beta1 ~ dnorm(0, 1/5^2)                             # Slope on elevation

  # Joint likelihood: Note identical alpha and beta for all data sets
  # Likelihood portion for data set 1: regular counts
  for (i in 1:nsites1){
    C1[i] ~ dpois(lambda1[i])
    log(lambda1[i]) <- beta0 + beta1 * selev1[i]
  }
  # Likelihood portion for data set 2: zero-truncated counts
  for (j in 1:nsites2){
    C2[j] ~ dpois(lambda2[j])T(1,)
    log(lambda2[j]) <- beta0 + beta1 * selev2[j]
  }
  # Likelihood portion for data set 3: detection/nondetection
  for (k in 1:nsites3){
    y[k] ~ dbern(psi[k])
    cloglog(psi[k]) <- beta0 + beta1 * selev3[k]
  }
}
```